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Meaningful Learning
8/26/2006

Having designed computer systems and enterprise software for the past 15 years, I can relate computer-based modeling tools, or rather technology-based modeling tools, to my work as a design engineer. Essentially, these modeling tools break a complex problem into small, manageable parts or, rather, explainable parts. Two of the conceptual change theory models—cognitive conflict theory and revisionist theory—essentially take a complex problem and break it down into many relatable learning activities—relations, operations, and interactions. In my field of engineering, this is not a new concept, and in fact, Universal Modeling Language (UML) is one of the many modeling tools that are used to assist engineers in problem solving or learning activities. Like any other mind tool, it is a thinking tool that assists in problem-solving. I agree that these two models are rooted in constructivist theory. In addition, I liked how the author defines technology's role as communicating ideas to learners.

I am still questioning what constitutes "meaningful learning". I think that learners learn how to solve problems. According to the book, if you cannot build a model of what you are studying, then you don't understand what you are studying (pg. 4). I understand that meaningful learning involves both conceptual change and development and is based on one's cognitive processes and experiences. I'm a bit surprised that memorization is not considered meaningful learning. Going through life, a person experiences many things, and sometimes he/she just has to memorize them; that is how other people have done it, and he/she is going to do the same thing. Many of the things one encounters, whether through learning or memorizing, are considered experiences, and they can be used to resolve cognitive conflict. I conclude that meaningful learning can take many forms as long as it is applied to knowledge-building activities.

I am a design engineer, and so I want to say a few things about system engineering. A typical design includes an analysis phase and a design phase, and essentially, it's a problem-solving activity. Engineering techniques and tools are used to assist engineers in these activities—the effectiveness of our tools could be greatly increased if we based their design on scientific knowledge about how humans solve problems.

Just from reading the first two chapters, the book has some really good concepts, and it's not just for engineers like me; it's for anyone who uses computers to assist with their problem-solving. I think that technology-based modeling is a vital tool, because technology is such an important part of today's society. Overall, I liked the book, but I did not like the cognitive model and revisionist model diagrams because they're too confusing—with too many external factors pointing to a given situation. This kind of model is essentially a thinking model for "do this," then "do that," and if "do that" does not work, "try this." Generally speaking, people understand this kind of logic. I guess that this is why the author called his model "complex". I would not categorize this kind of thinking model as a theory; rather, it is just a thinking model (or a mind tool), because everyone can come up with a completely different model to solve the exact same problem. It's just a thinking tool.